

Process Analysis: Practice Set 2

I. ABC Contract Manufacturer

ABC is a contract manufacturer of critical sub-assemblies for XYZ mobile. It currently manufactures only a single type of sub-assembly exclusively for XYZ mobile. Table 1 lists the sequence of operations in the manufacturing of these sub-assemblies at ABC. ABC operates 1 shift (8 hours) a day, and needs to setup its machines every day it has to start production. ABC is a new start-up, and has of late been finding it difficult to meet the growing demand from XYZ mobile. ABC is now considering ramping up its capacity to be able to meet the growing demand from XYZ mobile.

- Where should it first increase capacity?
- Now suppose that operation C-1 produces 15% defective items, which are detected only after all the operations are over. All such defects need to be reworked at C-1 and all the subsequent operations. Rework does not produce any defect. At which operation should ABC first increase capacity if it needs to increase the overall capacity?
- Now suppose XYZ mobile is planning to launch a new model of its mobile, which needs a different subassembly for which the sequence of operations and the corresponding setup times and run times are given in Table 2. It is clear from Table 2 that the new subassembly does not need the C-2 operation. XYZ mobile has now placed an order for 50/day of the new sub-assembly along with 100/day of the older type. ABC is planning to produce the older and the newer sub-assembly types in batches of 100 and 50, respectively. Does ABC have enough capacity to meet these new demand requirements? If not, where should it consider adding capacity? Assume 15% defects for older sub-assembly type as described in (b), and no defects for newer sub-assembly type.

Operation	Setup time (min)	Run time/unit(min)
A	30	2
B	35	2.5
C		
C-1	45	1.5
C-2	20	2.3
C-3	50	1.25
D	0	1

Table 1: Setup and Run Times for Old Assembly

Operation	Setup time (min)	Run time/unit(min)
A	35	2.5
B	40	3.2
C		
C-1	50	2
C-2	0	0
C-3	60	1.4
D	0	1

Table 2: Setup and Run Times for New Assembly

II Service Line Redesign at RTO Ahmedabad

The Road Transport Office (RTO) Ahmedabad is under pressure to increase its productivity to accommodate 120 license renewal applications per hour with the addition of only one clerk to its present staff (5 clerks and an officer). Presently, the license renewal process is designed as a service (assembly!) line, with applicants being processed in the fixed serial sequence listed in Table below. Activity 1 (i.e. Review application for correctness) must be performed first, and activity 6 (i.e. issue renewed license) must be the last step and by Government policy be handled by a uniformed officer. Activities 4 (i.e. Conduct eye test) and 5 (i.e. Photograph applicant) require expensive eye testing machine and digital camera respectively.

Table: License Renewal Process Times

Activity	Description	Resource	Processing Time (Sec)
1	Review application for correctness	Clerk A	15
2	Process and record payment	Clerk B	30
3	Check for violations and restrictions	Clerk C	60
4	Conduct eye test	Clerk D	40
5	Photograph applicant	Clerk E	20
6	Issue renewed license	Officer	30

1. What is the maximum output possible per hour in the present system? What is the average direct labor utilization for the existing process?
2. Redesign the service delivery process keeping in mind the increased demand of renewal applications and availability of one additional clerk (than the present staffing level). Draw process flow diagram for your redesign. What is the direct labor utilization for your redesign?
3. What do you conclude from (1) and (2) above?
4. Revisit present design. Suppose, “Checking for violations and restrictions” is done on a computer terminal with new integrated database. This reduces the processing time of these checks to 30 seconds instead of 60 sections.
 - a. What is the bottleneck now? What is maximum output possible?
5. Suggest a reallocation of activities among 5 clerks and the officer that would result in a service capacity of 120 applicants per hour. What investment would be required to implement your reallocation? Can you improve the design so as to reduce the investment?

III. Indian Army

The pre-induction physical examination at Indian Army involves the following seven activities:

Activity	Average Time (Minutes)
Medical history	10
Blood Tests	8
Eye Examination	5
Measurements (i.e. weight, height, blood pressure)	7
Medical examination	16
Psychological interview	12
Exit medical evaluation	10

These activities can be performed in any order, with two exceptions: Medical history must be taken first, and exit medical evaluation is last. At present, there are three paramedics and two physicians on duty during each shift. Only physicians can perform exit evaluations and conduct psychological interviews. Other activities can be carried out by either physicians or paramedics.

- Develop a layout and balance the line.
- Which activity accounts for the current bottleneck?
- How many people can be processed per hour?
- What is the total idle time per cycle?
- If one more physician and one more paramedic can be placed on duty, how will you redraw the layout? What is the new throughput?

IV. Pizzeria WIMWIana

Given the poor quality of mess food, many WIMWI (Well-Known Institute of Management in Western India) students are regularly ordering expensive pizzas from the food chains of US Pizza, Domino's etc. You and your dormmate are contemplating to offer 100% fresh, customized, high quality pizza on a timely basis from the campus itself (now there is a space vacated by "Joos" too!). You know that the concessions (free space, storage shelves, and electricity!) given by the institute to the students for entrepreneurial ventures will also work in your favor. You have already picked some utensils and an oven too. You are now worrying about getting the operating system right. You have decided the following process flow.

Your dormmate would take orders, which you estimate would take about two minutes per order (irrespective of number of pizzas). An order can consist of just one pizza; or many pizzas of same or different types. In other words, orders would specify the types of pizza: thin crust versus deep dish, a variety of fillings and toppings, and quantity for each pizza type. Once the order is received, you would measure and mix the ingredients for the crust (flour, water, spices, etc.) in about five minutes per pizza type and then you would spoon the dough onto a pizza tray, which would take you another minute per pizza. Meanwhile, your dormmate would pick the order-specific filling and toppings (three minutes per pizza type) and clean and hash them (2 minutes per pizza). You would then pour the filling onto the dough and spread the toppings on the top (2 minutes per pizza), load and set the oven-timer (one minute). The actual baking would take about 11 min. The oven can accommodate only one pizza. During the next minute, your dormmate would unload the oven and put the pizza in a box. You would finish the process by accepting payment and handing over the box(es) containing fresh, warm pizza(s) to your customer. This takes one minute.

Questions:

- Draw a process flow chart identifying resources, activity times and any potential storage buffers.
- What is the minimum time that you can quote to your customer for an urgent order of 1 pizza?
- If Pizzeria WIMWIana operates only during 1800-2400 hrs, what is the capacity (assume all orders are of 1 pizza each)?
- What is the minimum time that you can quote to your customer for an order of 2 pizzas of same type? Should you give discount to customers asking for 2 pizzas of same type?
- What happens to the capacity (assuming all orders are of 1 pizza each), if you decide to run Pizzeria WIMWIana alone on your own i.e. without your dormmate?